Compiler vs Interpreter

Csa1402-Complier design

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I Introduction

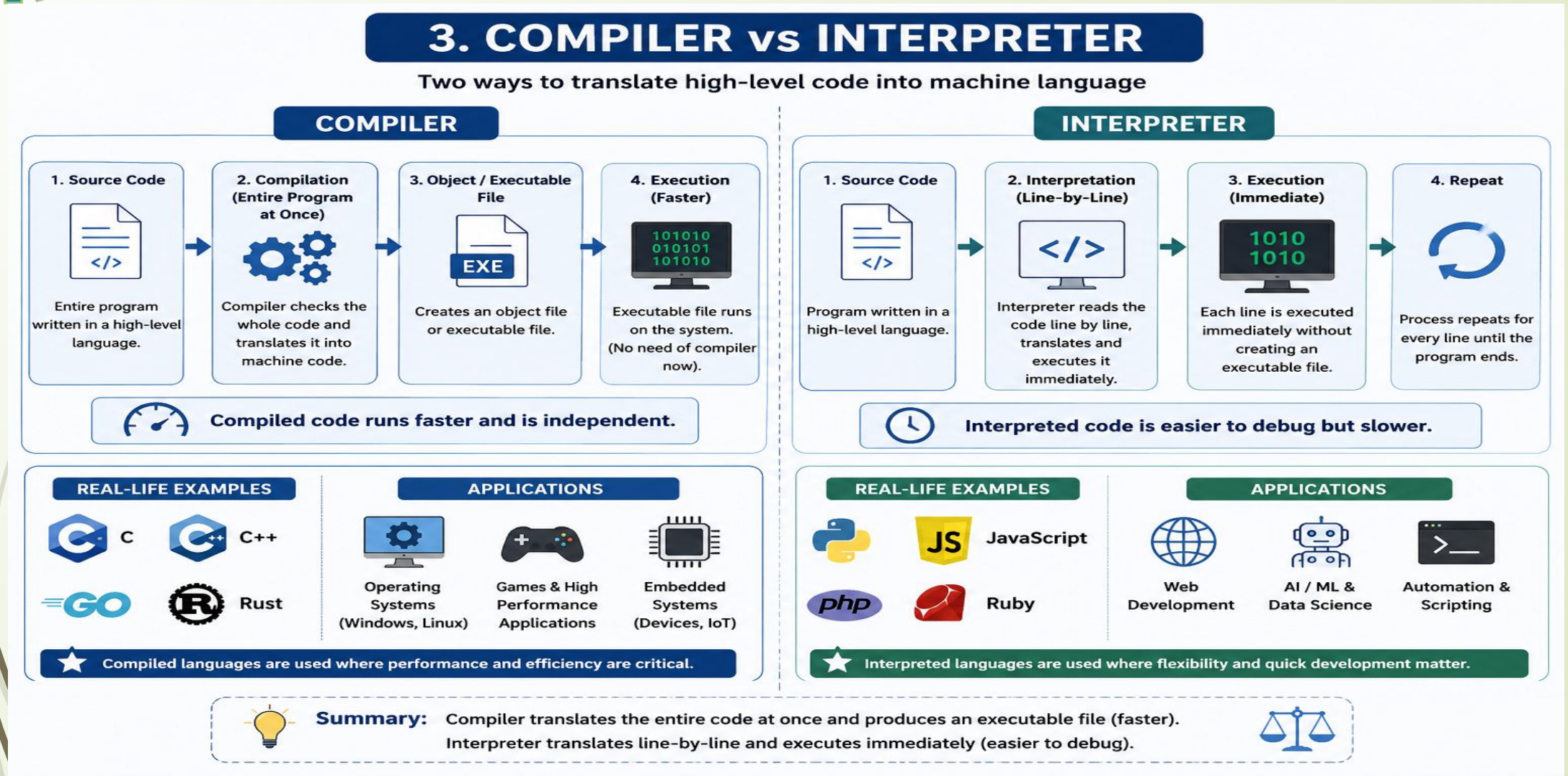
- **Compiler** and **Interpreter** are language translators used to convert high-level programming code into machine code.
- A **Compiler** translates the entire program before execution, producing an executable file.
- An **Interpreter** translates and executes the program one statement at a time.
- Both are essential for running programs, but they differ in speed, error handling, and execution process.
- Understanding these differences helps developers choose the appropriate translation method for different applications.

What is a Compiler?

- A **compiler** is a software program that converts the **entire source code** written in a high-level programming language into **machine code**.
- It translates the program **before execution** and creates an **executable file**.
- The compiled program can run directly on the computer without further translation.
- Compilers improve execution speed because the translation process is completed beforehand.
- **Examples:** C, C++, Go, Rust
- **Working Process**
- **Source Code → Compiler → Machine Code → Program Execution**

What is an Interpreter?

- An **interpreter** is a software program that converts high-level programming code into **machine code line by line**.
- It translates and executes each statement **immediately** without creating a separate executable file.
- Errors are displayed **as soon as they are encountered**, making debugging easier.
- Interpreted programs are generally slower because translation happens during execution.
- **Examples:** Python, JavaScript, Ruby
- **Working Process**
- **Source Code → Interpreter → Line-by-Line Translation & Execution**


3. COMPILER vs INTERPRETER
Two ways to translate high-level code into machine language
COMPILER
1. Source Code
Entire program written in a high-level language.
2. Compilation (Entire Program at Once)
Compiler checks the whole code and translates it into machine code.
3. Object / Executable File
EXE
Creates an object file or executable file.
4. Execution (Faster)
101010
010101
101010
Executable file runs on the system. (No need of compiler now).
Compiled code runs faster and is independent.
REAL-LIFE EXAMPLES
C
C++
Rust
APPLICATIONS
Operating Systems (Windows, Linux)
Games & High Performance Applications
Embedded Systems (Devices, IoT)
Compiled languages are used where performance and efficiency are critical.
INTERPRETER
1. Source Code
Program written in a high-level language.
2. Interpretation (Line-by-Line)
Interpreter reads the code line by line, translates and executes it immediately.
3. Execution (Immediate)
1010
1010
Each line is executed immediately without creating an executable file.
4. Repeat
Process repeats for every line until the program ends.
Interpreted code is easier to debug but slower.
REAL-LIFE EXAMPLES
JavaScript
Ruby
APPLICATIONS
Web Development
AI / ML & Data Science
Automation & Scripting
Interpreted languages are used where flexibility and quick development matter.
Summary: Compiler translates the entire code at once and produces an executable file (faster).
Interpreter translates line-by-line and executes immediately (easier to debug).

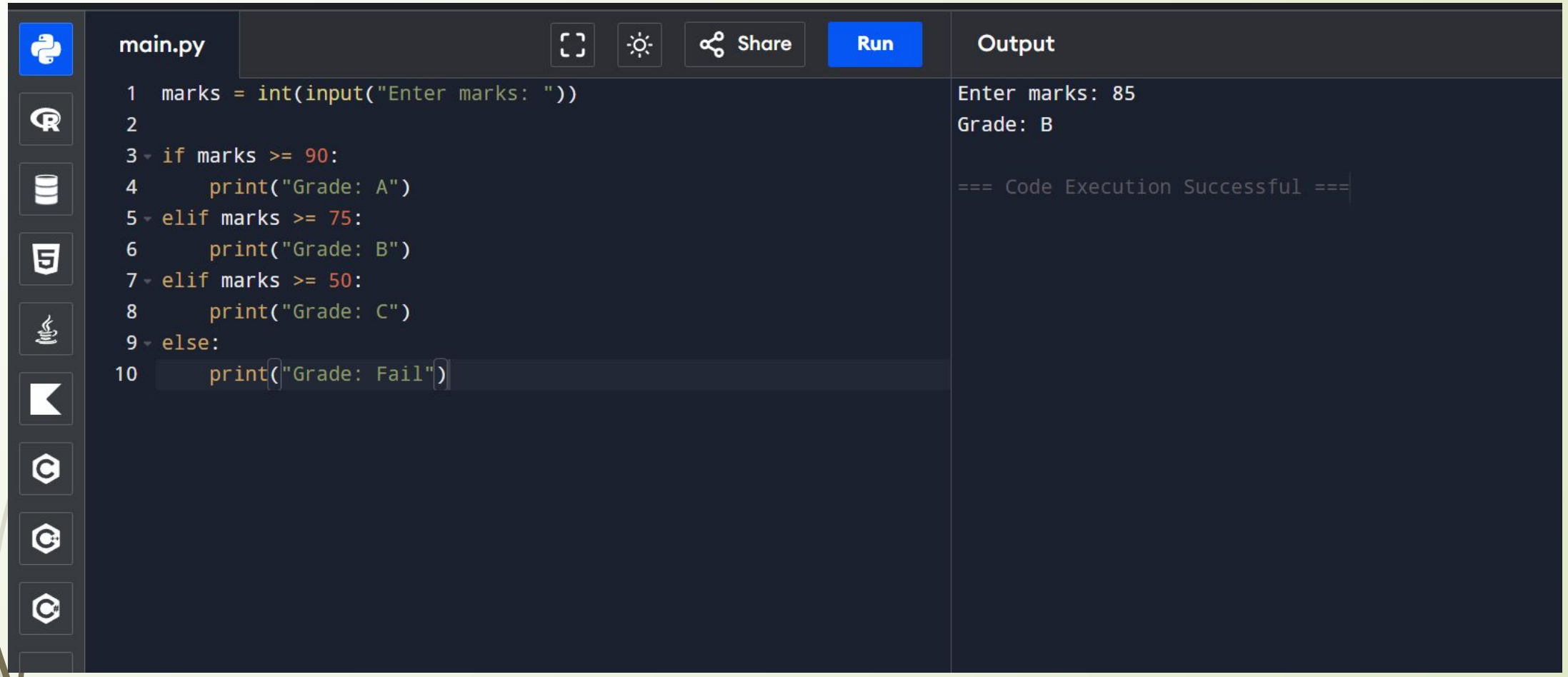
Engineer to Excel

Untitled1.cpp

```c
#include <stdio.h>

int main() {
    int marks;

    printf("Enter marks: ");
    scanf("%d", &marks);

    if (marks >= 90)
        printf("Grade: A");
    else if (marks >= 75)
        printf("Grade: B");
    else if (marks >= 50)
        printf("Grade: C");
    else
        printf("Grade: Fail");

    return 0;
}
```

C:\Users\sushm\OneDrive\Do

```
Enter marks: 85
Grade: B
--------------------------------
Process exited after 15.38 seconds with return value 0
Press any key to continue . . .
```


main.py
Share
Run
Output
1 marks = int(input("Enter marks: "))
2
3 if marks >= 90:
4 print("Grade: A")
5 elif marks >= 75:
6 print("Grade: B")
7 elif marks >= 50:
8 print("Grade: C")
9 else:
10 print("Grade: Fail")
Enter marks: 85
Grade: B
=== Code Execution Successful ===

How they works:

- **Compiler**
- Source Code (.cpp)
- |
- ▼
- Compiler (G++)
- (Translates Entire Program)
- |
- ▼
- Executable File (.exe)
- |
- ▼
- Program Execution
- |
- ▼
- Output: Grade = B

INTERPRETER

Source Code (.py)

|

▼

Python Interpreter
(Reads Line by Line)

|

▼

Executes Immediately

|

▼

Output: Grade = B

Conclusion

- Compiler translates the whole program at once.
- Interpreter executes the program line by line.
- Compiler is faster, while Interpreter is easier to debug.
- Both convert high-level code into machine language.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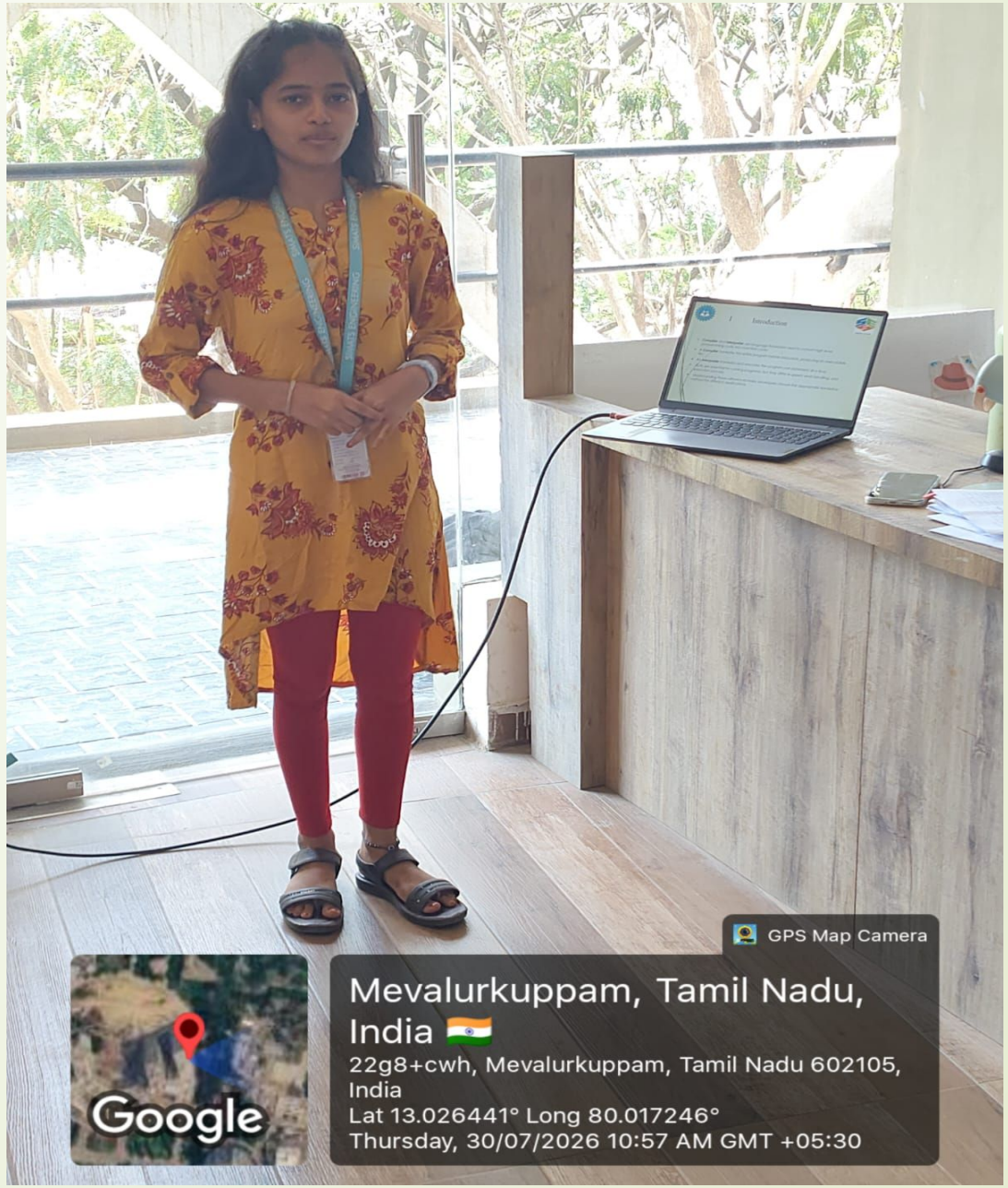
2005
Knowledge Conquers All


Engineer to Excel

Thank you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